Git Commands Cheat Sheet

A beginner's guide, in the order you'd actually use the commands

1. Setup (one time)

`git config --global user.name "Your Name"`	Set your name
`git config --global user.email "you@example.com"`	Set your email

2. Start a Repo

`git init`	Start a new repo in the current folder
`git clone <url>`	Copy an existing remote repo to your computer

3. Daily Workflow

`git status`	See what changed
`git add <file>`	Stage a file (git add . stages everything)
`git commit -m "message"`	Save staged changes with a message
`git log --oneline`	View short commit history
`git diff`	See unstaged changes

4. Connect to a Remote (GitHub, etc.)

`git remote -v`	Show where your remotes point
`git remote add origin <url>`	Link a remote to your local repo
`git remote remove <name>`	Delete a remote
`git remote rename <old> <new>`	Rename a remote

Pushing a cloned repo to a NEW remote URL (pick one)

First create an empty repo on GitHub (no README) and copy its URL.

`git remote set-url origin <NEW_URL>`	Option 1: replace the old repo with the new one (easiest)
`git remote add new-origin <NEW_URL>`	Option 2: keep the old repo and add the new one as a second remote
`git remote rename origin upstream`	Option 3, step 1: old repo becomes 'upstream'
`git remote add origin <NEW_URL>`	Option 3, step 2: new repo becomes 'origin'

After Option 1 or 3 push with `git push -u origin main`. After Option 2 push with `git push -u new-origin main`.

5. Push and Pull

`git push -u origin main`	First push of main (sets the default remote)
`git push`	Upload your commits
`git pull`	Download and merge remote changes
`git fetch`	Download changes without merging
`git push origin --all`	Push all branches
`git push origin --tags`	Push all tags

6. Branches (step by step)

The full flow for working on a feature in a separate branch:

1	`git branch`	List branches (the current one has a *)
2	`git switch -c <name>`	Create a new branch and switch to it
3	`git add . && git commit -m "msg"`	Work and commit as usual on the new branch
4	`git push -u origin <name>`	Push the NEW branch (first time only)
5	`git push`	Later pushes on this branch
6	`git switch main`	Go back to main
7	`git pull`	Get the latest main
8	`git merge <name>`	Merge your branch into main
9	`git push`	Push the updated main
10	`git branch -d <name>`	Delete the branch locally
11	`git push origin --delete <name>`	Delete the branch on the remote

Other branch commands

`git switch <name>`	Switch to an existing branch (older: git checkout <name>)
`git branch -a`	List local and remote branches
`git branch -m <new-name>`	Rename the current branch

7. Undoing Things

`git restore <file>`	Discard unstaged changes to a file
`git restore --staged <file>`	Unstage a file
`git commit --amend`	Edit the last commit
`git revert <commit>`	Safely undo a commit by adding a new one
`git reset --hard <commit>`	WARNING: discards everything after a commit (destructive)

8. Handy Extras

`git stash`	Shelve uncommitted work
`git stash pop`	Bring shelved work back
`git rm <file>`	Delete a file and stage the removal
`.gitignore`	A file listing paths Git should ignore

Start with: status, add, commit, push, pull. They cover most day-to-day use.